

```
[11]: import pandas as pd

[39]: df = pd.read_csv('/home/substramanyaprasad/Machine Learning/Data Analysis/Notes/03-Pandas/tips.csv')

[40]: df

[40]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251
...	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657
240	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	43752205509050	Sat17
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672

244 rows x 11 columns

```
[41]: df.head()
```

```
[41]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251

```
[42]: df.index
```

```
[42]: RangeIndex(start=0, stop=244, step=1)
```

We see that the index is normal, now let's change the index to their payment ID:

```
[43]: df.set_index('Payment ID')
```

```
[43]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
Payment ID										
Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
Sun4458	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322
Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
Sun2251	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221
...	...	...	...	...	...	...	...	...	...	...
Sat2657	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842
Sat1766	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404
Sat3880	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196
Sat17	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	43752205509050
Thur672	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139

244 rows x 10 columns

Now we can see that the index is now Payment ID and the column Payment ID at the end is removed

The Payment ID now is not a column, it's acutally the name of our index

But this change is not permanent:

```
[44]: df.head()
```

```
[44]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251

To make the changes permanent, follow the same way:

```
[45]: df.set_index('Payment ID',inplace=True)
```

```
[45]:
```

```
[46]: df.head()
```

```
[46]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
Payment ID										
Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
Sun4458	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322
Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
Sun2251	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221

Now to reset the index:

```
[47]: df.reset_index()
```

```
[47]:
```

	Payment ID	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
0	Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
1	Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
2	Sun4458	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322
3	Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
4	Sun2251	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221
...	...	...	...	...	...	...	...	...	...	...	...
239	Sat2657	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842
240	Sat1766	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404
241	Sat3880	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196
242	Sat17	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	43752205509050
243	Thur672	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139

244 rows x 11 columns

But this change is not permanent:

```
[48]: df.head()
```

```
[48]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
Payment ID										
Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
Sun4458	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322
Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
Sun2251	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221

```
[49]: df.reset_index(inplace=True)
```

```
[49]:
```

```
[50]: df.head()
```

```
[50]:
```

	Payment ID	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
0	Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
1	Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
2	Sun4458	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322
3	Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
4	Sun2251	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221

Now we see that the changes are permanent

```
[51]: df.set_index('Payment ID',inplace=True)
```

```
[51]:
```

Now, to grab things based on their location: (number)

```
[52]: df.iloc[0]
```

```
[52]:
```

total_bill	16.99
tip	1.01
sex	Female
smoker	No
day	Sun
time	Dinner
size	2
price_per_person	8.49
Payer Name	Christy Cunningham
CC Number	3560325168603410
Name:	Sun2959, dtype: object

We see that series is returned, now if we want by label:

```
[53]: df.loc['Sun2959']
```

```
[53]:
```

total_bill	16.99
tip	1.01
sex	Female
smoker	No
day	Sun
time	Dinner
size	2
price_per_person	8.49
Payer Name	Christy Cunningham
CC Number	3560325168603410
Name:	Sun2959, dtype: object

Suppose I want to grab rows 0 1 2 and 3: (same slicing as in normal python)

```
[54]: df.iloc[0:4]
```

```
[54]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
Payment ID										
Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
Sun4458	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322
Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994

```
[55]: df.loc(['Sun2959','Sun5260'])
```

```
[55]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
Payment ID										
Sun2959	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410
Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994

Now let's talk about removing rows and inserting new rows:

```
[56]: df.drop('Sun2959',axis=0,inplace=True)
```

```
[56]:
```

```
[34]: df
```

```
[34]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
0	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
1	21.01	3.5	Male	No	Sun	Dinner	3	7.0	Travis Walters	6011812112971322
2	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
3	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221
4	25.29	4.71	Male	No	Sun	Dinner	4	6.32	Erik Smith	213140353657882
...	...	...	...	...	...	...	...	...	...	...
240	22.67	2.0	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196
241	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	43752205509050
242	18.78	3.0	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139
243	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230

245 rows x 10 columns

Let's insert a new row:

```
[57]: one_row = df.iloc[0]
```

```
[57]:
```

total_bill	10.34
tip	1.66
sex	Male
smoker	No
day	Sun
time	Dinner
size	3
price_per_person	3.45
Payer Name	Douglas Tucker
CC Number	4478071379779230
Name:	Sun4608, dtype: object

```
[58]: df.append(one_row)
```

```
[58]:
```

```
AttributeError                                Traceback (most recent call last)
Cell In[58], line 1
----> 1 df.append(one_row)

File ~/Machine Learning/Data Analysis/venv/1365/python3.14/site-packages/pandas/core/generic.py:6286, in NDFrame._getattr__(self, name)
   6282     def name:
   6283         name not in self._accessors
   6284     self._info_axis._can_hold_identifiers_and_holds_name(name)
   6285     return self[name]
-> 6286     return object.__getattribute__(self, name)

AttributeError: 'DataFrame' object has no attribute 'append'
```

In new Pandas, this method got removed, instead use this:

If one_row is a DataFrame: df = pd.concat([df, one_row], ignore_index=True)

If one_row is a Series: df = pd.concat([df, one_row.to_frame().T], ignore_index=True)

If one_row is a dictionary: df = pd.concat([df, pd.DataFrame([one_row])], ignore_index=True)

To check what one_row is: type(one_row)

```
[59]: type(one_row)
```

```
[59]: pandas.Series
```

```
[60]: df = pd.concat([df, one_row.to_frame().T])
```

```
[60]:
```

```
[61]: df
```

```
[61]:
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230
Sun4458	21.01	3.5	Male	No	Sun	Dinner	3	7.0	Travis Walters	6011812112971322
Sun5260	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994
Sun2251	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221
Sun9079	25.29	4.71	Male	No	Sun	Dinner	4	6.32	Erik Smith	213140353657882
...	...	...	...	...	...	...	...	...	...	...
Sat1766	27.18	2.0	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404
Sat3880	22.67	2.0	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196
Sat17	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	43752205509050
Thur672	18.78	3.0	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139
Sun4608	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230

244 rows x 10 columns

We see that the first row, is appended at the very end